School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Project Scribe Report

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Scribe Question 1: Project System and Objective

What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Probabilistic Problem Being Addressed

The main probabilistic problem in our Device-to-Device (D2D) Edge Computing project is the uncertainty in **task processing time (TPT)**, meaning the exact time required to execute a task is not known beforehand. Most existing approaches assume this time is fixed or deterministic, but in real systems, task execution time varies depending on factors such as input data size and task complexity. Relying on an average execution time can lead to heavy or complex tasks taking much longer than expected, which may tie up resources, cause congestion, and delay other tasks.

Understanding the Problem (Highway Analogy)

Consider a highway toll booth: if the booth assumes every vehicle takes exactly 10 seconds to pass, the arrival of a large truck that takes 60 seconds will create a traffic jam for all cars behind it. Similarly, in our network, treating heavy tasks as normal can block the processing queue and delay other tasks. This demonstrates why a probabilistic approach is needed: the system should account for variability in task execution times instead of assuming fixed durations.

System Objective

The goal of our system is to **maximize the Task Completion Rate**, i.e., completing as many tasks as possible within their deadlines. We propose a **game-based probabilistic offloading approach** where each device estimates the likelihood of successfully completing a task before offloading it to another node. Using **game theory concepts**, particularly **Nash Equilibrium**, devices make balanced decisions to prevent any single node from being overloaded.

Primary Sources of Uncertainty

Task Processing Time (TPT):

- Execution time varies with task complexity and input size. For example, processing an image with many features takes longer than a simple image.

Queueing and Contention Delay:

- Multiple tasks competing for limited resources create unpredictable waiting times.

Device Heterogeneity:

- Different devices have varying computational power and workloads, leading to inconsistent execution times.

Transmission Delay:

- Offloading tasks to other devices or nodes introduces variable communication delays depending on network conditions.

Probabilistic Modeling Approach

To manage these uncertainties, particularly TPT variability, our system uses a **probabilistic model** (specifically, an exponential distribution). This allows for more realistic scheduling and smarter task offloading decisions.

System Illustration

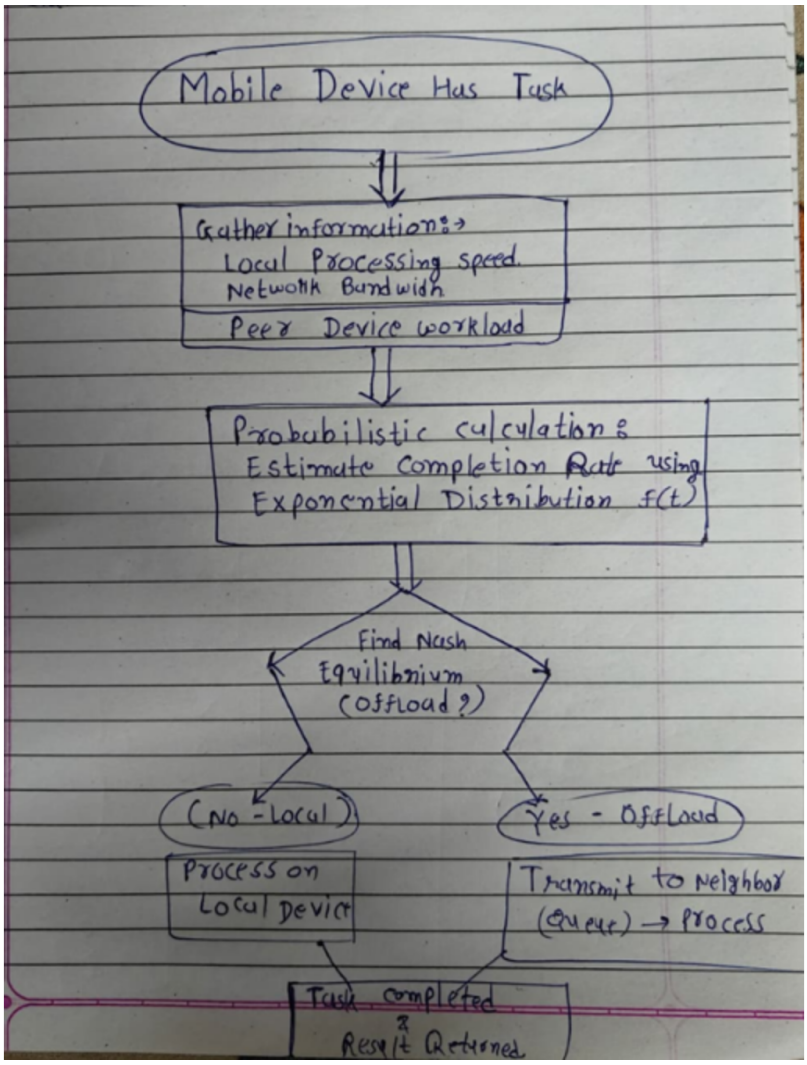

Figure 1: System model illustrating probabilistic task processing and offloading in D2D Edge Computing (Q1).

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Key Random Variables

Several aspects of the D2D Edge Computing system are treated as **random variables** because their exact values cannot be known in advance:

Task Processing Time (TPT):

- Modeled as a **continuous random variable** since it depends on the application type and input data size.
- Different applications behave differently, so application-specific probability distributions (e.g., exponential, Gaussian, Gamma) are used instead of a single fixed model.

Task Arrival Process:

- Modeled as a **Poisson process** with arrival rate λ_k.
- Captures the randomness of user-generated tasks, where exact arrival times are unpredictable.

Task Completion Time:

- Combines **queue waiting time, processing time, and transmission delay** (for offloaded tasks).
- Since each component is uncertain, total completion time is also a random variable.

Task Completion Indicator:

- Modeled as a **Bernoulli random variable**, taking value 1 if the task meets its deadline and 0 otherwise.
- Directly reflects system performance in terms of task success.

Device Workload (λ_j):

- Represents the arrival rate of tasks at a given device.
- Since arrivals fluctuate over time, the workload is dynamic and uncertain.

Uncertainty Modeling

- **TPT distributions:** Exponential, Gaussian, Gamma, or generalized extreme value distributions are used depending on the application behavior.
- **Offline profiling:** Historical execution data is used to select the most suitable distribution for each application type.
- **Online parameter calibration:** Distribution parameters are updated at runtime with a few samples to adapt to system changes while maintaining computational efficiency.

Current Probabilistic Assumptions

- **Independent task arrivals:** Arrival of one task does not affect others.
- **i.i.d. service times:** Tasks of the same type are assumed independent and identically distributed, allowing consistent probabilistic modeling.
- **Stable short-term dynamics:** System conditions are relatively stable during decision-making, so probability estimates are reliable.

Scribe Question 3: Probabilistic Reasoning and Dependencies

Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

In our project, **probabilistic reasoning** is central to how devices make task offloading decisions. The system explicitly models **dependence, conditionality, and independence** to capture interactions between devices and enable efficient decision-making.

Dependence Between Devices

Devices in a Device-to-Device (D2D) network are **interdependent** because they share computational resources.

- Offloading a task to another device increases the target's workload and queue length, affecting the probability that subsequent tasks assigned there will meet their deadlines.
- If multiple devices send heavy tasks to the same node, the task completion rate depends on the collective decisions of all devices.
- To manage this, our system uses **game-theoretic reasoning**, where each device accounts for congestion effects caused by others. Overall performance emerges from the collective behavior rather than individual independent actions.

Conditional Probability for Decision-Making

The offloading algorithm uses **conditional probability** to make informed decisions. Instead of relying on general device speed, it evaluates:

$$P(\text{Success} \mid \text{Target} = j)$$

where "Success" indicates that a task finishes before its deadline and j is the chosen offloading target.

- This probability depends on the current queue length, processing rate, and workload of the target device.
- By conditioning on the target device, the system avoids congested nodes and selects the most suitable target under current network conditions.
- Example: Processing a photo with 1 face is fast, while a photo with 20 faces takes longer. Probabilistic modeling (e.g., exponential distribution) estimates success probabilities accordingly.

Independence Assumptions for Simplified Analysis

To ensure computational efficiency in real-time decision-making, certain independence assumptions are made:

- **Independent task arrivals:** One task's arrival does not influence the next.
- **Memoryless service times:** Using an exponential distribution for TPT, the remaining processing time is independent of time already spent.

These assumptions simplify queueing analysis and allow efficient computation of completion probabilities using an **M/G/1 queueing model**, where task completion probabilities are derived from the **CDF** of processing time distributions rather than fixed thresholds.

Decision-Making and Nash Equilibrium

- Devices iteratively update offloading strategies considering others' decisions until a **Nash Equilibrium** is reached.
- At equilibrium, no device can improve its task completion probability by unilaterally changing its decision.

Dependency Diagram

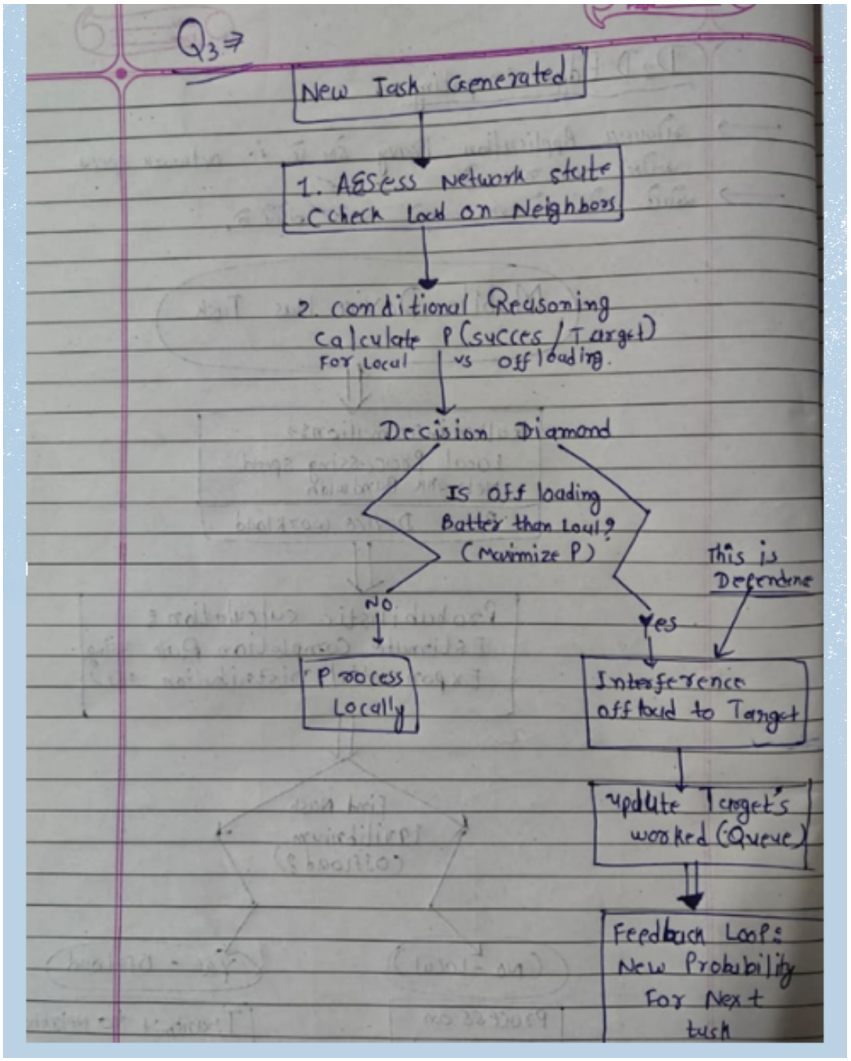

Figure 2: Probabilistic Dependencies and Conditional Reasoning between Devices (Q3).

Scribe Question 4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

At this stage, our group has mainly read the paper to understand how the probabilistic model is intended to work. The main idea is that task processing time (TPT) is uncertain, and probability distributions are used instead of fixed averages to model this uncertainty.

How the model connects to implementation and experiments in the paper

Task Processing Time (TPT) Model:

- Each application type is linked to a probability distribution (e.g., exponential for image tasks, Gamma or Gaussian for speech tasks).

Completion Rate Metric:

- The probability that a task finishes before its deadline is estimated using TPT distributions and M/G/1 queueing theory.
- This metric is both used by the offloading algorithm and measured during experiments.

Queueing Assumptions:

- Task arrivals follow a Poisson process.
- Each device processes tasks sequentially on a single CPU, matching the M/G/1 model.

Direct Use in Algorithm:

- The Mature offloading algorithm directly uses the probabilistic completion rate to decide where tasks should be executed.

Key Assumptions Affecting Design and Evaluation

Poisson Task Arrivals:

- Enables analytical modeling using M/G/1 queueing theory.

Single-Server Processing:

- Each device is modeled as a single-server system.

Known TPT Distributions:

- Distribution types and parameters are assumed to be learned beforehand.

Deadline-Oriented Metric:

- Completion rate is prioritized over average delay to suit time-sensitive applications.

Independent but Interacting Devices:

- Devices act independently, but task offloading decisions affect shared workloads.

Scribe Question 5: Cross-Milestone Consistency and Change

Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

At this stage, our group has mainly read and understood the paper. The probabilistic model provides a framework for handling uncertainty in Task Processing Time (TPT). Instead of assuming that every task takes a fixed amount of time, probability distributions are used to capture variability.

Current State of the Probabilistic Model

Task Processing Time (TPT) Model:

- Each type of application is modeled using a specific probability distribution—for example, exponential for image tasks and Gamma or Gaussian for speech tasks.

Completion Rate Metric:

- The model predicts the probability that a task finishes before its deadline.
- This aligns with the system's goal of deadline-aware task offloading.

Basic Assumptions:

- Tasks arrive according to a Poisson process.
- Each device processes tasks sequentially on a single CPU (M/G/1 assumption).
- Transmission times are considered deterministic.

Aspects Expected to Evolve in Future Milestones

Task Arrival Process:

- The Poisson assumption may be refined to handle bursty or non-Poisson arrivals.

TPT Distribution Selection:

- Current distributions are taken from the paper; future milestones may explore automatic or adaptive selection for new tasks.

Transmission and Communication Modeling:

- Deterministic transmission times may evolve into stochastic models to better reflect real network conditions.

Parallelism and Multi-Core Devices:

- The single-CPU assumption may be extended to multi-core devices with concurrent task execution.

Parameter Adaptation:

- Parameters such as service rate μ may require refinement to account for varying workloads and device conditions.

Justification

At this early stage (Milestone 1), our understanding is exploratory. The main framework of the model is clear: we understand why task times vary, how probability distributions are used, and how the completion rate is calculated. The expected changes above reflect the fact that real-world systems are more complex than our initial simplified assumptions. Future milestones will involve refined modeling, experiments, and analysis to improve alignment between the probabilistic model and actual system behavior.

Scribe Question 6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

At this stage, our understanding of the probabilistic aspects of the project is still developing. Several open issues and ambiguities remain.

Open Issues

Choice of Probability Distributions for Task Processing Time (TPT):

- Different applications use different distributions (e.g., exponential for image tasks, Gamma/Gaussian for speech tasks).
- A systematic method to select the appropriate distribution for a new or unknown application is not yet established, which introduces uncertainty in TPT predictions.

Reliability of Learned Parameters (μ):

- Parameters estimated from observed data may lose accuracy under changing system conditions.
- The frequency of parameter recalculation to maintain model reliability is currently unclear.

Poisson Task Arrival Assumption:

- Tasks are assumed to follow a Poisson arrival process.
- Real-world arrivals may deviate from this assumption, potentially affecting prediction accuracy.

Simplified Transmission Time Modeling:

- Transmission times are modeled as fixed, deterministic values.
- Real network conditions fluctuate, so deterministic modeling may oversimplify actual behavior.

Information Sharing Between Devices:

- The model assumes perfect sharing of workload and system information.
- In practice, information may be delayed or imperfect, impacting performance.

Responsibility Allocation for Next Milestone

Scribe Team (2 members):

- Document open issues, summarize observations, and record reasoning.

Video/PPT Team (2 members):

- Explain and clarify probabilistic concepts in the milestone video.

Concept Map/GitHub Member (1 member):

- Visualize relationships, assumptions, and probabilistic elements.
- Manage project files on GitHub.

Although each member has primary responsibilities, all team members actively contribute ideas, review content, and provide feedback across all deliverables. This collaborative approach ensures shared understanding and maintains the quality of our work.

The open issues identified will be addressed in future milestones through deeper modeling, analysis, experimentation, and enhanced visualization. Documenting these issues now and assigning responsibilities establishes a foundation for refining assumptions, resolving ambiguities, and strengthening the probabilistic model as the project progresses.

End of Scribe Report Submission